

5G RAN

Ultimate Video Experience Feature Parameter Description

Issue	Draft B
Date	2026-01-30



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Contents

1

Change History

1

2

About This Document

4

2.1

General Statements

4

2.2

Features in This Document

5

2.3

Differences Between NSA and SA

5

2.4

Differences Between High Frequency Bands and Low Frequency Bands

5

3

Overview

7

4

Intelligent Uplink Experience Guarantee (Trial)

8

4.1

Carrier Selection Based on Uplink Live Streaming Services

8

4.1.1

Principles

8

4.1.1.1

Live Streaming UE Identification

10

4.1.1.2

Triggering Carrier Selection Based on Uplink Live Streaming Services

10

4.1.1.3

Measurement Configuration Delivery

11

4.1.1.4

Measurement Configuration Report

12

4.1.1.5

User Experience Evaluation

13

4.1.1.6

Optimal Carrier or Carrier Combination Selection

15

4.1.1.7

Carrier Selection Based on Uplink Live Streaming Services

16

4.1.2

Network Analysis

17

4.1.2.1

Benefits

17

4.1.2.2

Impacts

17

4.1.3

Requirements

18

4.1.3.1

Licenses

18

4.1.3.2

Functions

18

4.1.3.2.1

Prerequisite Functions

18

4.1.3.2.2

Mutually Exclusive Functions

19

4.1.3.2.3

Function Impacts

19

4.1.3.3

Hardware

20

4.1.3.4

Networking

20

4.1.3.5

Others

20

4.1.4

Operation and Maintenance

21

4.1.4.1

Data Configuration

21

4.1.4.1.1

Data Preparation

21

4.1.4.1.2 Using MML Commands..... 22

4.1.4.1.3 Using the MAE-Deployment..... 23

4.1.4.2 Activation Verification..... 23

4.1.4.3 Network Monitoring..... 24

4.2 Dynamic Adjustment of Live Streaming Resources..... 24

4.2.1 Principles..... 24

4.2.2 Network Analysis..... 25

4.2.2.1 Benefits..... 25

4.2.2.2 Impacts..... 25

4.2.3 Requirements..... 25

4.2.3.1 Licenses..... 26

4.2.3.2 Functions..... 26

4.2.3.2.1 Prerequisite Functions..... 26

4.2.3.2.2 Mutually Exclusive Functions..... 27

4.2.3.2.3 Function Impacts..... 28

4.2.3.3 Hardware..... 28

4.2.3.4 Networking..... 29

4.2.3.5 Others..... 29

4.2.4 Operation and Maintenance..... 29

4.2.4.1 Data Configuration..... 29

4.2.4.1.1 Data Preparation..... 29

4.2.4.1.2 Using MML Commands..... 30

4.2.4.1.3 Using the MAE-Deployment..... 31

4.2.4.2 Activation Verification..... 31

4.2.4.3 Network Monitoring..... 31

5 Uplink Service Guarantee..... 32

5.1 Principles..... 32

5.1.1 Uplink Live Streaming UE Identification..... 32

5.1.2 Uplink Live Webcast Experience Guarantee..... 32

5.1.3 Uplink Differentiated Service Guarantee..... 34

5.1.4 Differentiated Uplink MU-MIMO Scheduling..... 34

5.2 Network Analysis..... 35

5.2.1 Benefits..... 35

5.2.2 Impacts..... 37

5.3 Requirements..... 39

5.3.1 Licenses..... 39

5.3.2 Functions..... 39

5.3.2.1 Prerequisite Functions..... 39

5.3.2.2 Mutually Exclusive Functions..... 41

5.3.2.3 Function Impacts..... 41

5.3.3 Hardware..... 41

5.3.4 Others..... 42

5.4 Operation and Maintenance..... 42

5.4.1 Data Configuration..... 42

5.4.1.1 Data Preparation..... 42

5.4.1.2 Using MML Commands..... 45

5.4.2 Using the MAE-Deployment..... 47

5.4.3 Activation Verification..... 47

5.4.4 Network Monitoring..... 48

6 Parameters..... 49

7 Counters..... 51

8 Glossary..... 52

9 Reference Documents..... 53

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 01 (2025-03-10).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between live streaming UE identification and time-domain power saving BWP. For details, see 4.1.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added differentiated uplink MU-MIMO scheduling. For details, see 5.1.4 Differentiated Uplink MU-MIMO Scheduling .	Added the NRDUCellULMimo.GrpPairSrRbCtrlCeFailThld parameter. Modified parameters: - Added the UL_MU_BIG_PKT_PAIR_RB_PO LI_SW and UL_MU_GRP_PAIR_RB_TILT_SW options to the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter. - Added the UL_MU_GRP_PAIR_SR_RB_CTRL_SW option to the NRDUCellULMimo.ULMuMimoAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the uplink differentiated service guarantee function. For details, see 5.1.3 Uplink Differentiated Service Guarantee .	Added parameters: - NRDUCellPusch.SrSchOptCceFailRateThld - NRDUCellPusch.SrSchOptPrbLowThld - NRDUCellPusch.SrSchOptPrbHighThld Modified parameters: - Added the SERVICE_DIFF_GUARANTEE_UL_FLAG option to the NRDUCellQciBearer.ServiceDiffAlgoSwitch parameter. - Added the UL_SERV_DIFF_EXP_GUAR_SW option to the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
- Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.
- Revised descriptions of the gain analysis and activation verification methods for uplink differentiated service guarantee. For details, see [5.2.1 Benefits](#) and [5.4.3 Activation Verification](#).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

Feature ID	Feature Name	Chapter/Section
FOFD-080019	Intelligent Uplink Experience Guarantee (Trial)	4 Intelligent Uplink Experience Guarantee (Trial)
FOFD-091253	Uplink Service Guarantee	5 Uplink Service Guarantee

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Intelligent Uplink Experience Guarantee (Trial)	Supported only in SA networking	4 Intelligent Uplink Experience Guarantee (Trial)
Uplink Service Guarantee	None	5 Uplink Service Guarantee

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high

frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Intelligent Uplink Experience Guarantee (Trial)	Supported only in low frequency bands	4 Intelligent Uplink Experience Guarantee (Trial)
Uplink Service Guarantee	Supported only in low frequency bands	5 Uplink Service Guarantee

3 Overview

Live streaming services feature heavy traffic in the uplink. To guarantee and improve the experience of such services, Huawei introduces the ultimate video experience solution.

4 Intelligent Uplink Experience Guarantee (Trial)

In multi-frequency scenarios, this feature is introduced to ensure user experience of uplink live streaming services and improve the service level agreement (SLA) capability. This feature includes [4.1 Carrier Selection Based on Uplink Live Streaming Services](#) and [4.2 Dynamic Adjustment of Live Streaming Resources](#).

4.1 Carrier Selection Based on Uplink Live Streaming Services

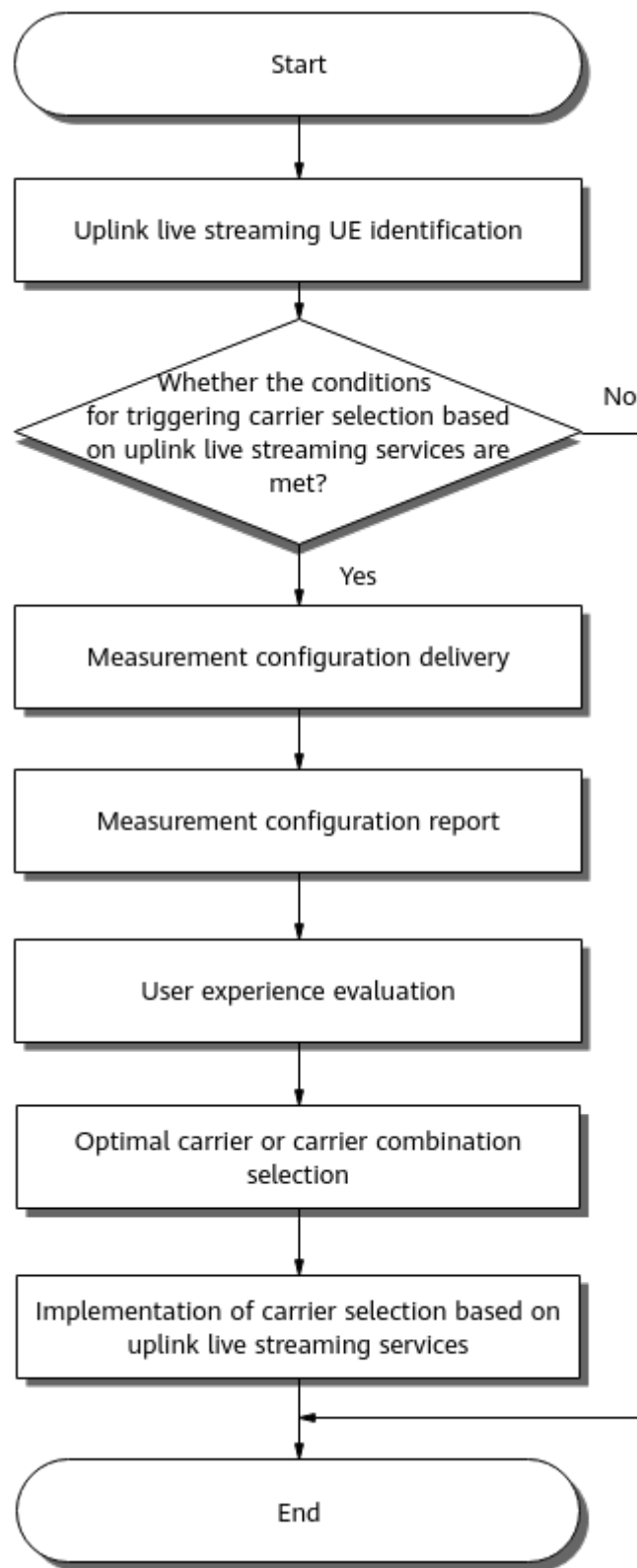
4.1.1 Principles

Carrier selection based on uplink live streaming services is jointly controlled by the **INTEL_UL_EXP_GUARANTEE_SW** option of the **NRCellFeatureSwitch.IntelligentExpGuarSwitch** parameter and the **UL_LIVE_SERVICE_FREQ_SEL_SW** option of the **NRCellAlgoSwitch.MultiFreqAlgoSwitch** parameter. After this function is enabled, if the uplink rate of live streaming services does not meet the rate indicated by the value of **NRDUCellQciBearer.ULLiveServGuarRate**, the gNodeB selects carriers or carrier combinations that provide the best uplink experience for live streaming UEs based on factors such as cell coverage, bandwidth, and load. [Figure 4-1](#) shows the basic process of this function.

Carrier selection based on uplink live streaming services requires the following configurations:

- The experience-based smart carrier selection function (controlled by the **MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch.MultiFreqAlgoSwitch** parameter) is enabled. For details about this function, see *Multi-Frequency Smart Selection*.
- **NRCellAlgoSwitch.MultiFreqSmartSelMode** is set to **UL_DL_MODE** or **UL_ONLY_MODE**.

Figure 4-1 Basic process of carrier selection based on uplink live streaming services



4.1.1.1 Live Streaming UE Identification describes the method of identifying live streaming UEs.

4.1.1.2 Triggering Carrier Selection Based on Uplink Live Streaming Services

describes the scenarios where carrier selection based on uplink live streaming services can be triggered.

4.1.1.3 Measurement Configuration Delivery describes how the gNodeB selects target frequencies for measurement.

4.1.1.4 Measurement Configuration Report describes measurement reports sent by UEs to the gNodeB.

4.1.1.5 User Experience Evaluation describes how the gNodeB evaluates user experience.

4.1.1.6 Optimal Carrier or Carrier Combination Selection describes the method for selecting an optimal carrier or carrier combination.

4.1.1.7 Carrier Selection Based on Uplink Live Streaming Services describes how the gNodeB determines whether to perform an inter-frequency handover or carrier change.

4.1.1.1 Live Streaming UE Identification

The NR system defines QoS bearers of different levels. Operators can use different QCI for live streaming UEs and other UEs and the gNodeB can identify live streaming UEs by QCI. Live streaming UEs must meet the following conditions:

- The UE has at least one bearer with a QCI for which the **gNBQciBearer.ServiceType** parameter is set to **EMBB_LIVE_VIDEO**.
- The UE is not running a VoNR service.
- If a live streaming UE accesses the network before the function of carrier selection based on uplink live streaming services is enabled, this function does not take effect for the UE.

4.1.1.2 Triggering Carrier Selection Based on Uplink Live Streaming Services

Carrier selection based on uplink live streaming services will be triggered when the triggering conditions of any triggering scenario are met, as listed in [Table 4-1](#).

Table 4-1 Triggering scenarios and conditions for carrier selection based on uplink live streaming services

Triggering Scenario	Triggering Condition
Rate status change of uplink live streaming services:	The uplink service rate of a live streaming UE changes from the guaranteed rate for uplink streaming services (specified by NRDUCellQciBearer.ULLiveServGuarRate) to a smaller value. (If the first triggering fails, there is another short-term guarantee of smart carrier selection.)
Traffic volume	The traffic volume of a live streaming UE (which is usually small during initial access) increases and the live streaming rate does not reach the guaranteed rate for uplink streaming services.

Triggering Scenario	Triggering Condition
Uplink weak coverage	- The gNodeB periodically (every 15s) checks the SRS SINR of the cell where the UE camps, and the value is less than the SRS SINR threshold (NRDUCellSrsMeas.NrToEutranSinrLowThld parameter value plus 8 dB). - The live streaming rate does not reach the guaranteed rate for uplink streaming services.
Periodic check	All of the following conditions are met: - The periodic timer specified by NRCeCellCaMgmtConfig.CarrSelMinInterval expires. - The uplink air interface capability of the UE decreases by more than 30% compared with that when multi-frequency smart selection takes effect last time. (For details about air interface capability evaluation, see 4.1.1.5 User Experience Evaluation). - The live streaming rate does not reach the guaranteed rate for uplink streaming services.

4.1.1.3 Measurement Configuration Delivery

The gNodeB can deliver measurement configurations for a maximum of four neighboring frequencies to a UE. When carrier selection based on uplink live streaming services is triggered for a UE meeting the QCI requirements, the gNodeB delivers inter-frequency event A5 measurement configurations related to the frequencies to be measured all together to the UE. (Threshold 1 and threshold 2 for event A5 are -43 dBm and -140 dBm, respectively.)

The principles based on which the gNodeB selects frequencies for inter-frequency measurement configuration delivery vary with UEs, as described in [Table 4-2](#).

Table 4-2 Frequency selection principles

UE Capability	Frequency Selection Principle
Single-carrier UEs or multi-carrier UEs supporting super uplink	- Selects and delivers the NR frequencies specified by the NRCeCellFreqRelation.SsbDescMethod and NRCeCellFreqRelation.SsbFreqPos parameters. - Selects and delivers a maximum of four neighboring frequencies based on the frequency priorities for RRC_CONNECTED UEs specified by the NRCeCellFreqRelation.ConnFreqPriority parameter.

UE Capability	Frequency Selection Principle
Multi-carrier UEs supporting CA	1. Preferentially selects and delivers the NR frequencies specified by the NRCellFreqRelation.SsbDescMethod and NRCellFreqRelation.SsbFreqPos parameters. 2. Additionally delivers other frequencies in the group of CA frequencies (configured in the gNBCaFrequency MOs) if fewer than four frequencies can be delivered in the previous step.

Before delivering measurement configurations to a UE, the gNodeB determines whether carrier selection based on uplink live streaming services can take effect for the UE based on the **gNBQciBearer.NetworkingOptionOptFlag** parameter settings corresponding to the QCLs of services running on the UE. The carrier selection policy varies depending on the parameter value, as described in [Table 4-3](#).

Table 4-3 Policy for carrier selection based on uplink live streaming services to take effect

gNBQciBearer.NetworkingOptionOptFlag Parameter Value for the QCLs of Services Running on a UE	gNBQciBearer.NetworkingOptionOptFlag Parameter Value for the QCLs of Services Running on Other UEs	Whether Carrier Selection Based on Uplink Live Streaming Services Takes Effect on the UE
PERMIT	None	Yes
PERMIT	FORBID	No
PERMIT	PERMIT or FOLLOW	Yes
FOLLOW	None	No
FOLLOW	FOLLOW or FORBID	No
FOLLOW	PERMIT	Yes
FORBID	None	No
FORBID	PERMIT, FOLLOW, or FORBID	No

4.1.1.4 Measurement Configuration Report

A UE performs measurements based on the event measurement configuration and reports measurement results.

- If the UE sends measurement reports for all neighboring frequencies contained in the measurement configurations within the time that the

gNodeB waits for inter-frequency event A5 measurement reports, user experience evaluation will start. For details about user experience evaluation, see [4.1.1.5 User Experience Evaluation](#).

- If the time that the gNodeB waits for inter-frequency event A5 measurement reports has ended but no cells on some neighboring frequencies measured by the UE meet conditions, the gNodeB deletes the inter-frequency event A5 measurement configurations for these neighboring frequencies so that the UE stops the current inter-frequency event A5 measurements. Then, user experience evaluation is started for the UE. For details, see [4.1.1.5 User Experience Evaluation](#).

 NOTE

The time that the gNodeB waits for inter-frequency A5 measurement reports is related to the number of measured neighboring frequencies.

4.1.1.5 User Experience Evaluation

The gNodeB selects cells based on the received inter-frequency event A5 measurement reports. That is, it selects the cells that meet the conditions below from the cells each with the highest RSRP value on the corresponding neighboring frequencies. The gNodeB evaluates user experience in the serving cell and in these selected cells.

First, cells that can be candidate carriers or can be added to a candidate carrier combination will be selected.

- If the RSRP of the cell with the largest RSRP value on a neighboring frequency is greater than or equal to the RSRP threshold for interference-based inter-frequency handovers (**NRCeIIInterFHoMeaGrp.IntrfInterFreqHoRsrpThld**), this cell can be selected as a candidate carrier or as a primary component carrier (PCC) in a candidate carrier combination for user experience evaluation.
- If the RSRP of the cell with the largest RSRP value on a neighboring frequency is greater than or equal to the event A5 threshold 2 (equal to the **NRCeIIaMgmtConfig.CaA5RsrpThld2** parameter value plus the **gNBCaFrequency.CaScCA5RsrpThld2Offset** parameter value) for secondary cell (SCell) configuration for CA, the cell can be selected as a secondary component carrier (SCC) in a candidate carrier combination for user experience evaluation.
- A cell failing to meet the preceding conditions will not be selected for user experience evaluation.

Then, user experience evaluation is performed for candidate carriers or cells in candidate carrier combinations in the following aspects:

Table 4-4 Experience evaluation method

Evaluation Dimension		UE Type	Evaluation Method
Uplink air interface capability	UE-specific uplink spectral efficiency	Uplink-preferred UEs	The uplink spectral efficiency of a UE in an inter-frequency candidate cell is calculated based on the uplink spectral efficiency of the UE in its serving cell and the differences in the path loss, interference, and number of uplink receive antennas between the serving and candidate cells.
	Cell-specific number of remaining uplink RBs		This number is calculated based on the accumulated number of remaining RBs on the PUSCH per second in the cell involved.
Uplink air interface capability		Multi-carrier UE	The uplink air interface capability of a candidate carrier combination is equal to the sum of the uplink air interface capabilities of all carriers in the combination.
Equivalent cell bandwidth		N/A	This bandwidth is calculated based on factors such as the cell bandwidth ^a , slot configuration ^c , and UE-supported bandwidth ^b .
CCE allocation failure rate		N/A	This rate is calculated based on the total number of CCE allocation times and the number of successful CCE allocation times. ^d
a: The uplink cell bandwidth is specified by the NRDUCell.UlBandwidth parameter. b: The UE-supported bandwidth can be obtained from the <i>channelBWs-UL</i> and <i>channelBWs-UL-v1590</i> IEs in the <i>UECapabilityInformation</i> IE signaled over the Uu interface. For details, see section 4.2 "UE Capability Parameters" in 3GPP TS 38.306 V16.5.0. c: The slot configuration is specified by the NRDUCell.SlotAssignment parameter. d: CCE allocation failures affect uplink service experience of UEs. If the uplink CCE allocation failure rate exceeds 20%, the corresponding candidate carrier or carrier combination will not be considered for optimal carrier or carrier combination selection.			

 NOTE

- UEs cannot measure the SSB beams of independently deployed super uplink cells. Therefore, an independently deployed super uplink cell can be a candidate for user experience evaluation for a UE only if it is the current serving cell of the UE. For details about independently deployed super uplink cells, see *Super Uplink*.
- The gNodeB evaluates user experience based on the measurement reports received from UEs. The evaluation can be affected by factors such as measurement report accuracy, gNodeB configuration differences, and UE compatibility differences. Therefore, the average uplink UE throughputs after handovers triggered by carrier selection based on uplink live streaming services may not be as expected. When carrier selection based on uplink live streaming services is used on the live network, it is recommended that the parameter **NRCeMgmtConfig.UExpEvalHyst** be used for optimization. The larger the values of these parameters, the smaller the impact on the average uplink UE throughputs after handovers triggered by experience-based smart carrier selection. The smaller the values of these parameters, the larger the impact on the average uplink throughputs after handovers triggered by experience-based smart carrier selection.
- The differences in the path loss and interference between the serving and candidate cells affect uplink service experience of UEs. The impact of the differences on user experience can be reduced by setting the **NRCeMgmtConfig.UExpEvalHyst** or **NRDUCellCarrMgmt.UCellExpEvaluationFactor** parameter on the gNodeB.
- The difference in the number of uplink receive antennas between cells has an impact on uplink service experience of UEs. The **NRDUCellCarrMgmt.UCellExpEvaluationFactor** parameter can be used on the gNodeB to configure a factor for uplink experience evaluation in individual cells based on antenna TX/RX modes (for example, 4T4R) of the cells.
- Inter-gNodeB CA affects uplink service experience of UEs due to inter-gNodeB transmission delay. The impacts can be reduced by setting the **NRCeMgmtConfig.InterGnbExpEvalCoeff** parameter, which specifies the inter-gNodeB experience evaluation coefficient for individual cells.

4.1.1.6 Optimal Carrier or Carrier Combination Selection

The gNodeB selects an optimal carrier or carrier combination based on the user experience evaluation results (including the air interface capability and equivalent cell bandwidths), whether a handover or carrier change is involved, and the number of carriers. For an uplink-preferred UE, the factors listed in [Table 4-5](#) are evaluated in sequence. For example, if the candidates have the same uplink air interface capability, handover evaluation continues. The selection process ends when any factor is evaluated in a candidate's favor. This candidate is then regarded as the optimal carrier or carrier combination.

Table 4-5 Evaluation principles

Seq uen ce	Factor	Evaluation Principle
1	Air interface capability	Candidate carriers or carrier combinations with a higher air interface capability have a higher priority. An air interface capability hysteresis is used to compare the uplink air interface capability of a candidate carrier or carrier combination with that of the current carrier or carrier combination. If the difference in the uplink air interface capability between the two carriers or carrier combinations is less than the corresponding hysteresis, the carriers or carrier combinations are considered to have the same uplink air interface capability. The hysteresis for uplink air interface capability is specified by the NRCeIlCaMgmtCon-fig.UlExpEvalHyst parameter.
2	Handover ^a	A handover transiently interrupts data transmission. Therefore, candidate carrier combinations that do not involve a handover has a higher priority.
3	Carrier change ^a	A carrier change transiently interrupts data transmission. Therefore, candidate carrier combinations that do not involve an SCC change have a higher priority.
4	Equivalent cell bandwidth	Candidate carriers or carrier combinations with a larger equivalent cell bandwidth have a higher priority.
5	Number of carriers ^a	In the case of multi-carrier UEs, the smaller the number of carriers for aggregation in a candidate carrier combination, the lower the consumption of channel resources such as PUCCH resources. Therefore, candidate carrier combinations with a smaller number of carriers have a higher priority.
a: This factor needs to be considered only in optimal carrier combination selection for multi-carrier UEs.		

4.1.1.7 Carrier Selection Based on Uplink Live Streaming Services

If the selected optimal carrier or carrier combination is different from the current carrier or carrier combination, a handover or carrier change is required, as listed in [Table 4-6](#).

Table 4-6 Handover modes for different UEs

UE Capability	Whether the PCC in the Optimal Carrier Combination Is the Current PCC	Handover Mode
Single-carrier UE	N/A	A handover to the target cell is initiated directly.
Multi-carrier UE that supports CA or super uplink	No	A handover to the target cell needs to be initiated. During the handover, information of the SCCs or supplementary uplink (SUL) carriers ^a in the optimal carrier combination is delivered.
	Yes	The gNodeB sends an RRCReconfiguration message to change the SCC or SUL carrier ^a .
a: For details about SUL carriers, see <i>Super Uplink</i> .		

4.1.2 Network Analysis

4.1.2.1 Benefits

After this function is enabled, the gNodeB selects carriers or carrier combinations that provide the best uplink experience for uplink live streaming UEs based on factors such as spectral efficiency, bandwidth, and load. This guarantees more UEs whose rates meet the guaranteed rate (**NRDUCellQciBearer.UlliveServGuarRate**) of uplink live streaming services. This function brings greater benefits to networks with more frequencies, larger coverage difference between frequency bands, or larger load difference between cells.

It is recommended that the function of carrier selection based on uplink live streaming services be enabled at the network level. If this function is enabled for a single cell, the maximum average uplink UE throughput may not be obtained.

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

Carrier selection based on uplink live streaming services has the following impacts on the network:

- Inter-band UE distribution changes. The number of signaling messages over the air interface and the number of handovers increase.
- Frequent user experience evaluation increases the CPU usage of the main control board.

4.1.3 Requirements

4.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-080019	Intelligent Uplink Experience Guarantee (Trial)	NR0S00NUEG00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCCellAlgoSwitch.MultiFreqReqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	Carrier selection based on uplink live streaming services can be enabled only when experience-based smart carrier selection is enabled and NRCCellAlgoSwitch.MultiFreqSmartSelMode is set to UL_DL_MODE or UL_ONLY_MODE .
Low-frequency TDD	Multi-Frequency Smart Selection Mode	NRCCellAlgoSwitch.MultiFreqSmartSelMode	None	

4.1.3.2.2 Mutually Exclusive Functions

None

4.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch . MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	After carrier selection based on uplink live streaming services is enabled, smart carrier selection is no longer triggered for live streaming UEs.
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving . UePowerSavingSwitch parameter	<i>UE Power Saving</i>	Live streaming UEs do not switch to a sparse SSSG.
Low-frequency TDD	RedCap SSSG switching	REDCAP_SSSG_SWITCHING_SW option of the NRDUCellUePwrSaving . UePowerSavingSwitch parameter	<i>RedCap</i>	Live streaming UEs do not switch to a sparse SSSG.
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving . BwpPwrSavingSw parameter	<i>UE Power Saving</i>	Live streaming UEs do not switch to BWP2.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.3.4 Networking

- Only SA networking is supported.
- The network uses at least two frequencies.
- The Xn interface between gNodeBs is normal so that user experience evaluation information can be transmitted.

4.1.3.5 Others

- During key event assurance, it is recommended that carrier selection based on uplink live streaming services be disabled. This is to prevent network performance deterioration as the function increases CPU usage.
- If carrier selection based on uplink live streaming services takes effect in a single cell, UEs in this cell can be proactively handed over to inter-frequency cells and this function does not need to be enabled in candidate inter-frequency cells. If the gNodeB ID of an inter-frequency cell is different from that of the serving cell of a UE, it is recommended that experience-based smart carrier selection be enabled for at least one cell under the gNodeB serving the inter-frequency cell (for details, see *Multi-Frequency Smart Selection*). Otherwise, this inter-frequency cell cannot serve as a candidate carrier or as a PCC in a candidate carrier combination.
- If a cell with carrier selection based on uplink live streaming services disabled is activated first, for example, when its serving base station is reset, the cell cannot serve as a candidate carrier or as a PCC in a candidate carrier combination for user experience evaluation within 20 minutes after the activation.
- It is recommended that the number of frequencies configured on the gNodeB be greater than the maximum number of carriers supported by multi-carrier

UEs for CA. This will ensure that the gNodeB selects an optimal carrier combination for such UEs. The gNodeB can select an optimal carrier or carrier combination for a UE when the number of frequencies configured on the gNodeB is equal to the maximum number of carriers supported by the UE for CA in the following situations:

- If carrier selection based on uplink live streaming services takes effect and there are multi-carrier UEs supporting CA on the network, ensure that the basic functions of CA normally take effect.

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

[Table 4-7](#) describes the parameters used for function activation. For details about other related parameters, see [Table 4-8](#). This section does not describe parameters related to cell establishment.

Table 4-7 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Intelligent Experience Guarantee Switch	NRCellFeatureSwitch.IntelligentExpGuarSwitch	It is recommended that the INTEL_UL_EXP_GUARANTEE_SW option be selected to enable the Intelligent Uplink Experience Guarantee feature.
Low-frequency TDD	Multi-Frequency Algorithm Switch	NRCellAlgorithmSwitch.MultiFreqAlgoSwitch	It is recommended that the UL_LIVE_SERVICE_FREQ_SEL_SW option be selected to enable carrier selection based on uplink live streaming services.
Low-frequency TDD	Service Type	gNBQciBearer.ServiceType	The value EMBB_LIVE_VIDEO is recommended.
Low-frequency TDD	Uplink Live Service Guaranteed Rate	NRDUCellQciBearer.ULLiveServGuarRate	Set this parameter based on operators' requirements. It is not recommended that this parameter be set to 0.

Table 4-8 Other related parameters

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Multi-Frequency Algorithm Switch	NRCCellAlgoSwitch.MultiFreqAlgoSwitch	It is recommended that the MULTI_FREQ_SMART_SEL_SW option be selected to enable experience-based smart carrier selection.
Low-frequency TDD	Multi-Frequency Smart Selection Mode	NRCCellAlgoSwitch.MultiFreqSmartSelMode	The value UL_DL_MODE or UL_ONLY_MODE is recommended.
Low-frequency TDD	NR Move to E-UTRAN UL SINR Low Threshold	NRDUCellSrsMeas.NrToEutranSinrLowThld	Set this parameter to its recommended value.
Low-frequency TDD	Carrier Selection Minimum Interval	NRCCellCaMgmtConfig.CarSelMinInterval	Set this parameter to its recommended value.
Low-frequency TDD	NR 5G QoS Indicator	gNB5qiConfig.Nr5qi	Set this parameter based on operators' requirements.
Low-frequency TDD	QoS Class Identifier	gNB5qiConfig.Qci	Set this parameter based on operators' requirements.
Low-frequency TDD	Networking Option Opt Flag	gNBQciBearer.NetworkingOptionOptFlag	It is recommended that this parameter be set to PERMIT , indicating that carrier selection based on uplink live streaming services is allowed.

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The extended QCI of 128 is used as an example.

```
//Enabling Intelligent Uplink Experience Guarantee
MOD NRCELLFEATURESWITCH: NrCellId=0, IntelligentExpGuarSwitch=INTEL_UL_EXP_GUARANTEE_SW-1;
//Turning on the smart carrier selection switch, setting the mode of smart carrier selection (the
UL_DL_MODE is used as an example), and turning on the carrier selection switch for uplink live streaming
services
MOD NRCELLALGOSWITCH: NrCellId=0,
MultiFreqAlgoSwitch=MULTI_FREQ_SMART_SEL_SW-1&UL_LIVE_SERVICE_FREQ_SEL_SW-1,
MultiFreqSmartSelMode=UL_DL_MODE;
//Adding a QCI-specific bearer for the gNodeB, setting the service type to live streaming services,
configuring the Networking Option Opt Flag to allow carrier selection based on uplink live streaming
services
ADD GNBQCIBEARER: Qci=128, ServiceType=EMBB_LIVE_VIDEO, NetworkingOptionOptFlag=PERMIT;
//Adding 5QI configuration for the gNodeB
ADD GNB5QICONFIG: Nr5qi=128, Qci=128;
//Configuring bearers for the NR DU cell and the uplink live streaming service rate to be guaranteed
ADD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=128, ULLiveServGuarRate=8000;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the carrier selection switch for uplink live streaming services
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=UL_LIVE_SERVICE_FREQ_SEL_SW-0;
//Disabling smart carrier selection
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=MULTI_FREQ_SMART_SEL_SW-0;
//Disabling Intelligent Uplink Experience Guarantee
MOD NRCELLFEATURESWITCH: NrCellId=0, IntelligentExpGuarSwitch=INTEL_UL_EXP_GUARANTEE_SW-0;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

If the values of the counters (in [Table 4-9](#)) corresponding to 5QIs whose service type is **EMBB_LIVE_VIDEO** are not 0, the function of carrier selection based on uplink live streaming services has taken effect.

Table 4-9 Counters related to carrier selection based on uplink live streaming services

Counter ID	Counter Name
1911844800	N.Service.Exp.Guar.CS.Exec.Cell5QI

4.1.4.3 Network Monitoring

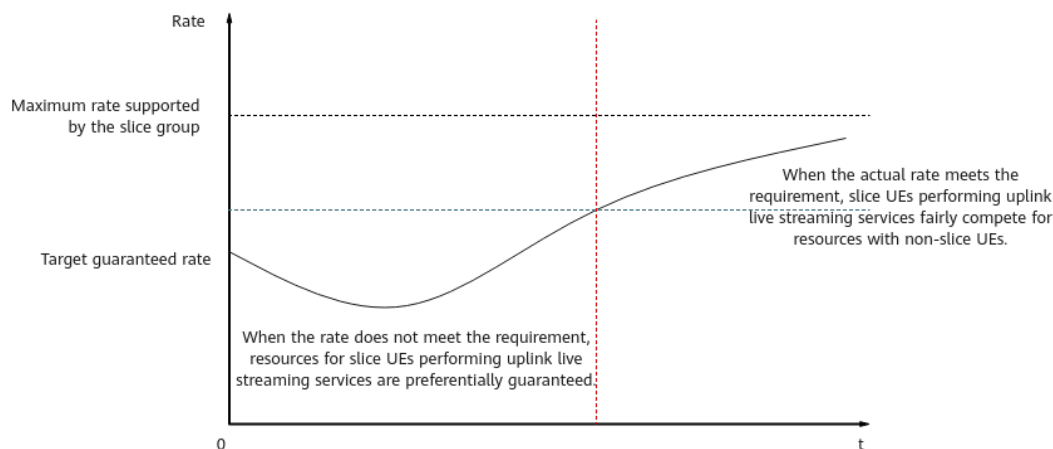
Observe the uplink RLC throughput of live streaming UEs to check whether the live streaming rate meets the guaranteed rate after this function is enabled.

4.2 Dynamic Adjustment of Live Streaming Resources

4.2.1 Principles

When network slices are used to guarantee uplink live streaming rates of sports events or live news, the uplink live streaming rate requirement cannot be met in real time because fixed RB resources are reserved for network slices. To address this issue, Huawei proposes the function of dynamic adjustment of live streaming resources. This function dynamically adjusts the preferential RB resources of a slice group to ensure second-level guarantee for rates of slice UEs performing uplink live streaming services. This feature is jointly controlled by the **INTEL_UL_EXP_GUARANTEE_SW** option of the **NRCellFeatureSwitch.IntelligentExpGuarSwitch** parameter and the **UL_RB_AVG_RATIO_DYN_ADJ_SW** option of the **NRDUCellRes.RbRatioDynAdjSwitch** parameter. [Figure 4-2](#) shows the rate adjustment.

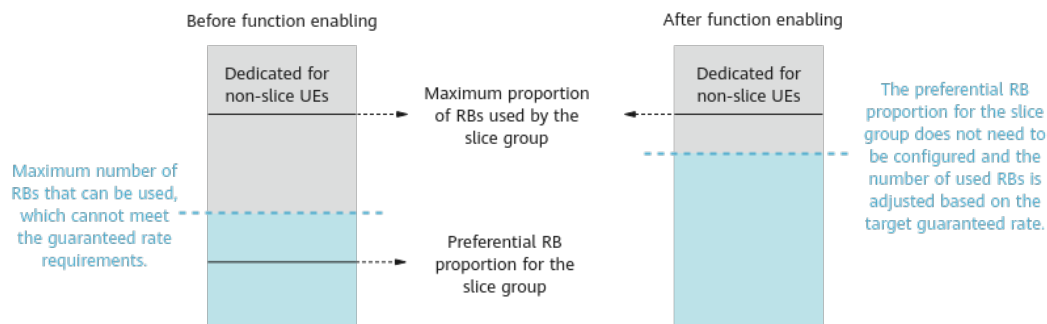
Figure 4-2 Rate adjustment



When the **INTEL_UL_EXP_GUARANTEE_SW** option of the **NRCellFeatureSwitch.IntelligentExpGuarSwitch** parameter and the **UL_RB_AVG_RATIO_DYN_ADJ_SW** option of the **NRDUCellRes.RbRatioDynAdjSwitch** parameter are selected, this feature dynamically adjusts the preferential spectrum resources allocated to uplink live streaming UEs based on air interface environment changes. When the rates of slice UEs performing uplink live streaming services do not meet the guaranteed rate, the spectrum resources of the live streaming UEs are preferentially guaranteed. When the rates of slice UEs performing uplink live streaming services meet the guaranteed rate, they fairly compete for spectrum resources with non-slice UEs. The uplink guaranteed rate for slice UEs performing uplink live streaming services is configured by the **gNBDUMacParamGroup.ULGuaranteedRate** parameter based on the operators'

requirements. [Figure 4-3](#) shows RB quantity adjustment during preferential guarantee.

Figure 4-3 RB quantity adjustment during preferential guarantee



4.2.2 Network Analysis

4.2.2.1 Benefits

After dynamic adjustment of live streaming resources is enabled (by selecting the **UL_RB_AVG_RATIO_DYN_ADJ_SW** option of the **NRDUCellRes.RbRatioDynAdjSwitch** parameter), the average uplink throughput of live streaming UEs in the network slice group configured with the uplink guaranteed rate increases.

It is recommended that this function be enabled when you cannot determine whether the reserved slice resources can meet the guaranteed rate of uplink live streaming services. If the maximum rate supported by the slice group is greater than the guaranteed rate of uplink live streaming services, then the rate requirements of uplink live streaming services can be met.

Note that when the network load is light, this function does not provide significant gains because resources are sufficient.

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

After dynamic adjustment of live streaming resources is enabled, the RB resources allocated to background service UEs decrease because live streaming UEs configured with the uplink guaranteed rate are preferentially guaranteed. As a result, the user-perceived rate of background service UEs may decrease.

4.2.3 Requirements

4.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-080019	Intelligent Uplink Experience Guarantee (Trial)	NR0S00NUEG00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference	Description
Lo w-freq uen cy TD D	Dynamic RB control	RB_DYNAMIC_CONTROL_SW option of the NRDUCell AlgoSwitch. <i>Network Slice AlgoSwitch</i> parameter	<i>Network Slicing</i>	This function depends on the basic functions of network slicing.

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Network Slicing Introducti on	NRDUCell Res.<i>ResUs ageObject</i>	<i>Network Slicing</i>	If the UL_RB_AVG_RATIO_DYN_ADJ_SW option of the NRDUCellRes.<i>RbRatioDynAdjSwit ch</i> parameter is selected, the NRDUCellRes.<i>ResUsageObject</i> parameter can only be set to NETWORK_SLICE_GROUP and the NRDUCellRes.<i>ResStrategyType</i> parameter can only be set to MEAN_ONLY .
Lo w- freq uen cy TD D	Network Slicing Introducti on	NRDUCell Res.<i>ResSt rategyTyp e</i>	<i>Network Slicing</i>	
Lo w- freq uen cy TD D	Network Slicing Introducti on	NRDUCell Res.<i>UlRbA verageRat io</i>	<i>Network Slicing</i>	If the UL_RB_AVG_RATIO_DYN_ADJ_SW option of the NRDUCellRes.<i>RbRatioDynAdjSwit ch</i> parameter is selected, the NRDUCellRes.<i>UlRbAverageRatio</i> parameter can only be set to 255 .

4.2.3.2.2 Mutually Exclusive Functions

None

4.2.3.2.3 Function Impacts

RA T	Function Name	Function Switch	Referenc e	Description
Lo w- freq uen cy TD D	Uplink experience -based scheduling	UL_EXP_BASED_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	When the uplink guaranteed rate and uplink experience-based scheduling functions take effect for different UEs, resources are preferentially allocated to the UE with the uplink guaranteed rate. When the uplink guaranteed rate and uplink experience-based scheduling simultaneously take effect for a UE, the UE rate is increased mainly based on the uplink guaranteed rate.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Networking

None

4.2.3.5 Others

Only live streaming equipment (such as LiveU equipment) is supported.

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Table 4-10 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-10 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Intelligent Experience Guarantee Switch	NRCellFeatureSwitch. <i>IntelligentExpGuarSwitch</i>	It is recommended that the INTEL_UL_EXP_GUARANTEE_SW option be selected to enable the Intelligent Uplink Experience Guarantee feature.
Low-frequency TDD	RB Ratio Dynamic Adjustment Switch	NRDUCellRes. <i>RbRatioDynAdjSwitch</i>	It is recommended that the UL_RB_AVG_RATIO_DYN_ADJ_SW option be selected to enable dynamic adjustment of live streaming resources.
Low-frequency TDD	NR 5G QoS Indicator	gNB5qiConfig. <i>Nr5qi</i>	Set this parameter based on operators' requirements.
Low-frequency TDD	QoS Class Identifier	gNB5qiConfig. <i>Qci</i>	Set this parameter based on operators' requirements.
Low-frequency TDD	MAC Parameter Group ID	gNBDUMacParameterGroup. <i>MacParamGroupId</i>	Set this parameter based on the QCI.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Uplink Guaranteed Rate	gNBDUMacParameterGroup.ULGuaranteedRate	Set this parameter based on operators' requirements. It is not recommended that this parameter be set to 0 for uplink services.

NOTE

- If live streaming equipment uses fixed bit rates, you are advised to set the uplink guaranteed rates based on the rates used by the equipment.
- If live streaming equipment uses variable rates, you are advised to set the uplink guaranteed rates based on the actual video quality requirements.
- The actual rates depend on the rate configuration of live streaming equipment.
- The modification of the uplink guaranteed rate parameter does not take effect immediately. Instead, it takes effect immediately after the UE re-accesses the network.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Take extended QCI 128 as an example. Before enabling dynamic adjustment of live streaming resources, you need to deploy network slicing. For details about how to deploy network slicing, see *Network Slicing*.

- Enabling Intelligent Uplink Experience Guarantee
 MOD NRCELLFEATURESWITCH: NrCellId=0,
 IntelligentExpGuarSwitch=INTEL_UL_EXP_GUARANTEE_SW-1;
- Configuring dynamic adjustment of live streaming resources
 //Configuring dynamic adjustment of live streaming resources
 ADD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=NETWORK_SLICE_GROUP,
 ResStrategyType=MEAN_ONLY, UIRbMaxRatio=70, UIRbAverageRatio=255,
 RbRatioDynAdjSwitch=UL_RB_AVG_RATIO_DYN_ADJ_SW-1;
- Configuring the uplink guaranteed rate for live streaming services
 //Adding QCI-specific bearer configuration for the gNodeB
 ADD GNBQCIBEARER: Qci=128;
 //Adding 5QI configuration for the gNodeB
 ADD GNB5QICONFIG: Nr5qi=128, Qci=128;
 //Adding a QCI-specific bearer for an NR DU cell and specifying the MAC parameter group ID and RLC parameter group ID referenced by the bearer

```
ADD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=128, MacParamGroupId=128, AmRlcParamGroupId=3,
UmRlcParamGroupId=0;
//Adding a QCI-specific bearer for the NR cell and the PDCP parameter group ID referenced by the
bearer
ADD NRCELLQCIBEARER: NrCellId=7, Qci=128, AmPdcPParamGroupId=5, UmPdcPParamGroupId=1;
//Adding an operator-level bearer with the QCI for the NR cell and the PDCP parameter group ID
referenced by the bearer
ADD NRCELLOPQCIBEARER: NrCellId=7, OperatorId=0, Qci=128, RlcMode=AM,
AmPdcPParamGroupId=5, UmPdcPParamGroupId=1;
//Configuring the uplink guaranteed rate
MOD GNBUMACPARAMGROUP: MacParamGroupId=128, UlGuaranteedRate=8000;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Deleting the bearer corresponding to the uplink live streaming services

```
//Removing QCI-specific bearer configuration from an NR cell
RMV NRCELLQCIBEARER: NrCellId=7, Qci=128;
//Removing operator-level bearer configuration for the QCI from an NR cell
RMV NRCELLOPQCIBEARER: NrCellId=7, OperatorId=0, Qci=128;
//Removing QCI-specific bearer configuration from an NR DU cell
RMV NRDUCELLQCIBEARER: NrDuCellId=7, Qci=128;
//Removing 5QI configuration from the gNodeB
RMV GNB5QICONFIG: Nr5qi=128;
//Removing QCI-specific bearer configuration from the gNodeB
RMV GNBQCIBEARER: Qci=128;
```

- Disabling dynamic adjustment of live streaming resources

```
//Turning off the switch for dynamic adjustment of live streaming resources
MOD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=NETWORK_SLICE_GROUP,
RbRatioDynAdjSwitch=UL_RB_AVG_RATIO_DYN_ADJ_SW-0;
//Disabling Intelligent Uplink Experience Guarantee
MOD NRCELLFEATURESWITCH: NrCellId=0,
IntelligentExpGuarSwitch=INTEL_UL_EXP_GUARANTEE_SW-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

Observe the changes in the uplink RLC throughput to check whether this feature has taken effect.

4.2.4.3 Network Monitoring

Check whether the uplink live streaming rate meets the requirement based on the uplink RLC throughput.

5 Uplink Service Guarantee

5.1 Principles

When there are numerous uplink live streaming UEs in a cell, a scheduling policy that does not distinguish UEs' priorities can easily lead to multiple UEs experiencing frame freezing simultaneously. Consequently, the target uplink rate cannot be achieved. Therefore, the Uplink Service Guarantee feature is introduced. This feature includes the [5.1.1 Uplink Live Streaming UE Identification](#), [5.1.2 Uplink Live Webcast Experience Guarantee](#), and [5.1.3 Uplink Differentiated Service Guarantee](#) functions.

5.1.1 Uplink Live Streaming UE Identification

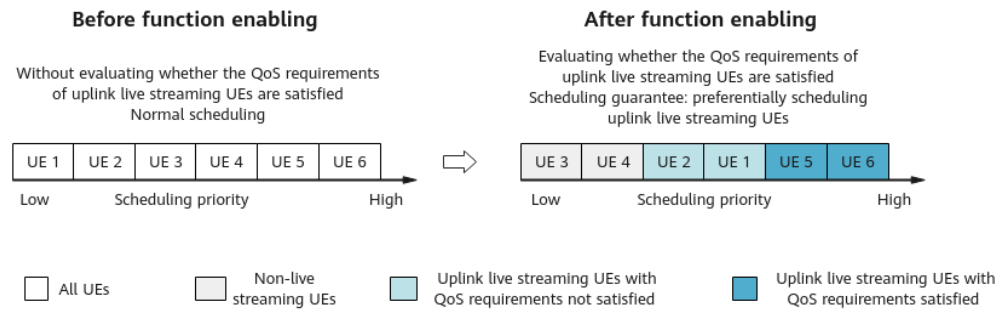
Operators can set different QCI to distinguish uplink live streaming UEs from other UEs. During RAB setup, the core network notifies the gNodeB of the 5QIs for different UEs. After receiving the 5QIs, the gNodeB maps them to the corresponding QCIs and identifies uplink live streaming UEs. The service type corresponding to a QCI is specified by the **gNBQciBearer.ServiceType** parameter. If this parameter is set to **UL_LIVE_WEBCAST_SERVICE**, a bearer with this QCI is used for uplink live streaming services.

5.1.2 Uplink Live Webcast Experience Guarantee

The uplink live webcast experience guarantee function ensures the uplink transmission rate of UEs through queuing mechanism and scheduling priority optimization, thereby improving user experience of uplink live streaming services. The principles of the involved technologies are as follows:

- Queuing mechanism: Uplink live streaming UEs are classified into two types, UEs with QoS requirements satisfied and UEs with QoS requirements not satisfied. The two queues are dynamically maintained, and the UEs with QoS requirements satisfied are preferentially guaranteed for constant QoS compliance.
- Scheduling priority: The scheduling priorities of uplink live streaming UEs are raised. In addition to UEs with QoS requirements satisfied, those with QoS requirements not satisfied are also preferentially scheduled. This ensures the uplink transmission rate of more uplink live streaming UEs in the cell.

Figure 5-1 Uplink live webcast experience guarantee



To ensure the uplink transmission rate of non-live streaming UEs and prevent uplink live streaming UEs at the cell edge that do not meet the QoS requirements from occupying too many RB resources, scheduling is not guaranteed for uplink live streaming UEs at the cell edge. For an uplink live streaming UE at the cell edge, its uplink SINR is less than the threshold specified by the **gNBDUSchParamGroup.UlliveWebcastFarUeSnrThld** parameter.

This function is controlled by the **UL_LIVE_WEBCAST_EXP_GUAR_SW** option of the **NRDUCellFeatureSw.QciBasedDiffServExpSw** parameter.

The uplink live webcast experience guarantee function takes effect at the UE level. Therefore, the following configurations are also required:

- Set the **gNBQciBearer.ServiceType** parameter to **UL_LIVE_WEBCAST_SERVICE**.
- Configure the **NRDUCellULSch.StableUIExpMaxUeNum** parameter as required.
- Set the **gNBDUSchParamGroup.StableUIExpGuarRate** parameter based on the uplink rate requirements of uplink live streaming UEs.
- Configure the **gNBDUSchParamGroup.StableUIExpPol** parameter as required:
 - It is recommended that this parameter be set to **EXPERIENCE_FIRST** if the rates of some uplink live streaming UEs need to be preferentially guaranteed.
 - It is recommended that this parameter be set to **USER_NUM_FIRST** when the capacity in terms of the number of uplink live streaming UEs needs to be preferentially guaranteed.
 - It is recommended that this parameter be set to **NEUTRAL** when both the rates and the capacity need to be guaranteed.

NOTE

Assume that multiple bearers (including bearers of the uplink live streaming services and other services) are set up for a UE. If the bearers of different QCI are configured with different priorities, parameters for the uplink live webcast experience guarantee function corresponding to the highest-bearer-priority QCI take effect for the UE. Therefore, to make this function take effect, for uplink live streaming UEs, ensure that the bearer priority of the uplink live streaming service is only lower than that of QCI 1. The priority is specified by the **gNBQciBearer.PriorityLevel** parameter. A smaller value indicates a higher priority.

RB Policy for Large-Packet Scheduling in Uplink MU-MIMO Pairing

In medium- and heavy-load scenarios with uplink scheduling enhancement enabled, the RB resource size available for pairing may not match the size required by UEs enabled with differentiated service guarantee during uplink MU pairing, resulting in pairing failure. To improve the experience of those UEs, "RB policy for large-packet scheduling in uplink MU-MIMO pairing" is introduced. After this function is enabled, the base station evaluates the RB resource size available for pairing, the RB resource size required by UEs enabled with differentiated service guarantee, and the remaining RB resource size of the cell, and determines whether to perform MU-MIMO pairing.

This function is controlled by the **UL_MU_BIG_PKT_PAIR_RB_POLI_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter.

RB Control for SR-based Scheduling in Uplink MU-MIMO Grouping and Pairing

In medium- or heavy-load scenarios with uplink MU grouping and pairing enabled, common UEs may occupy a large number of RB resources during SR-based scheduling. As a result, the resources available for UEs enabled with differentiated service guarantee decrease. To improve the experience of those UEs, "RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing" is introduced. After this function is enabled, the RB resources occupied by common UEs during SR-based scheduling are reduced.

This function is controlled by the **UL_MU_GRP_PAIR_RB_TILT_SW** option of the **NRDUCellQciBearer.ServiceDiffAlgoSwitch** parameter and the **UL_MU_GRP_PAIR_SR_RB_CTRL_SW** option of the **NRDUCellUIMimo.UlMuMimoAlgoSwitch** parameter.

This function takes effect only when the uplink CCE allocation failure rate of the cell is less than or equal to the **NRDUCellUIMimo.GrpPairSrRbCtrlCceFailThld** parameter value.

NOTE

- For details about uplink scheduling enhancement, see "PUSCH Time-Frequency Resource Allocation" in *Uplink Scheduling*.
- For TDD: For details about uplink MU grouping and pairing, see "Traffic Channel Interference Reduction" in *Massive MIMO Uplink Boosting (Low-Frequency TDD)*.

5.2 Network Analysis

5.2.1 Benefits

- Uplink live webcast experience guarantee
When the uplink live webcast experience guarantee function is enabled and all of the following conditions are met, the proportion of guarantee duration for uplink live streaming services in a cell increases by 5% to 20%, and the number of uplink live streaming UEs with QoS requirements satisfied increases by 10% to 20%. The gains vary depending on the actual situation of uplink live streaming services. If the following conditions are not fully met,

the gains provided by the uplink live webcast experience guarantee function decrease or are not significant: In NSA networking, the gains slightly decrease.

Table 5-1 Conditions for gains of uplink live webcast experience guarantee

Requirement	Description
UEs	Uplink live streaming UEs are evenly distributed at the cell center (RSRP greater than -75 dBm), a medium distance from the cell center (RSRP ranging from -85 dBm to -75 dBm), and cell edge (RSRP ranging from -100 dBm to -85 dBm).
Cells	- Uplink cell load ($(N.PRB.PUSCH.Used.Avg/N.PRB.UL.Avail.Avg \times 100) > 70\%$) - Number of activated non-live streaming UEs in the uplink ($N.User.RRConn.Active.DuCell5QI.Avg > 0$) - Proportion of non-live streaming service traffic ($(N.ThpVol.UL - N.ThpVol.UL.DuCell5QI)/N.ThpVol.UL > 10\%$) - Proportion of guarantee duration for live streaming services in a cell ($(N.Cell.Reserve.48 + N.Cell.Reserve.49 + N.Cell.Reserve.50)/(N.Cell.Reserve.41 + N.Cell.Reserve.42 + N.Cell.Reserve.43 + N.Cell.Reserve.44 + N.Cell.Reserve.45 + N.Cell.Reserve.46 + N.Cell.Reserve.47 + N.Cell.Reserve.48 + N.Cell.Reserve.49 + N.Cell.Reserve.50) < 70\%$)
Services	- Dedicated QCI configured for uplink live streaming services - The resolution of live streaming services is relatively consistent. For example, the proportion of 720p videos is greater than 90%.

- Uplink differentiated service guarantee
After uplink differentiated service guarantee is enabled, the average uplink rates of 5QI- and QCI-specific services increase if (1) the PRB usage and uplink CCE allocation failure rate meet the related conditions and (2) the minimum rate guarantee for non-GBR services or GBR services takes effect. The optimal gains can be achieved when the following functions are also used:
 - QCI-level configuration for power saving BWP
 - QCI-specific uplink power control optimization
 - QCI-specific downlink power control optimization
 - QCI-specific PDCCH aggregation level optimization
 - Differentiated gap-assisted measurement

NOTE

- For details about minimum rate guarantee for GBR services and minimum rate guarantee for non-GBR services, see *QoS Management*.
- For details about QCI-level configuration for power saving BWP, see "Frequency Domain" in *UE Power Saving*.
- For details about QCI-specific uplink power control optimization, QCI-specific downlink power control optimization, QCI-specific PDCCH aggregation level optimization, and differentiated gap-assisted measurement, see "Service-differentiated Link Reliability" in *Service-differentiated Experience Guarantee*.
- Average uplink rate of 5QI-specific services = $(N.ThpVolForRate.UL.DuCell5QI - N.ThpVolForRate.UL.SmallPkt.DuCell5QI) / N.ThpTimeForRate.UL.RmvSmallPkt.DuCell5QI$
- Average uplink rate of QCI-specific services = $(N.ThpVolForRate.UL.QCI - N.ThpVolForRate.UL.SmallPkt.QCI) / N.ThpTimeForRate.UL.RmvSmallPkt.QCI$
- Differentiated uplink MU-MIMO scheduling
After differentiated uplink MU-MIMO scheduling is enabled in medium- and heavy-load scenarios, the 5QI- and QCI-specific average uplink rates of UEs with specific QCIs increase, the number of RBs paired for uplink MU-MIMO in the cell increases, and both the total and average numbers of layers paired for uplink MU-MIMO increase.
 - Average uplink rate of 5QI-specific services = $(N.ThpVolForRate.UL.DuCell5QI - N.ThpVolForRate.UL.SmallPkt.DuCell5QI) / N.ThpTimeForRate.UL.RmvSmallPkt.DuCell5QI$
 - Average uplink rate of QCI-specific services = $(N.ThpVolForRate.UL.QCI - N.ThpVolForRate.UL.SmallPkt.QCI) / N.ThpTimeForRate.UL.RmvSmallPkt.QCI$
 - Number of RBs paired for uplink MU-MIMO: **N.ChMeas.MIMO.UL.Pair.PRB**
 - Total number of layers paired for uplink MU-MIMO: **N.ChMeas.MIMO.UL.Pair.Layer**
 - Average number of layers paired for uplink MU-MIMO: **N.ChMeas.MIMO.UL.Pair.Layer.Avg**

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

The uplink live webcast experience guarantee function has the following impacts:

- UEs whose data rates meet QoS requirements occupy a large number of uplink resources. As a result, the resources available for other UEs decrease. In uplink heavy-load scenarios ($(N.PR.B.PUSCH.Used.Avg / N.PR.B.UL.Avail.Avg) \times 100\% > 70\%$), the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) may decrease, and user experience of some uplink live streaming UEs at the cell edge also deteriorates.
- Changes in UE scheduling behavior and measurements of the number of UEs at different locations lead to fluctuations in the average uplink UE throughput, UE throughput distribution at the RLC layer, average uplink MCS index, uplink IBLER, and uplink RBLER. This, in turn, causes fluctuations in the

proportion of uplink small packets. The specific fluctuation depends on the UE distribution.

Table 5-2 lists the related indicators.

Table 5-2 Indicators affected by uplink live webcast experience guarantee

Indicator	Formula
Average uplink UE throughput	User Uplink Average Throughput (DU)
UE throughput distribution at the RLC layer	N.Thp.UL.Samp.Index0 to N.Thp.UL.Samp.Index11
Average uplink MCS index	$\frac{(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + 3 \times \text{N.ChMeas.PUSCH.MCS.3} + \dots + 27 \times \text{N.ChMeas.PUSCH.MCS.27} + 28 \times \text{N.ChMeas.PUSCH.MCS.28})}{(\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \text{N.ChMeas.PUSCH.MCS.3} + \dots + \text{N.ChMeas.PUSCH.MCS.27} + \text{N.ChMeas.PUSCH.MCS.28})}$
Uplink IBLER	$\frac{(\text{N.UL.SCH.HalfPiBPSK.ErrTB.Ibler} + \text{N.UL.SCH.QPSK.ErrTB.Ibler} + \text{N.UL.SCH.16QAM.ErrTB.Ibler} + \text{N.UL.SCH.64QAM.ErrTB.Ibler} + \text{N.UL.SCH.256QAM.ErrTB.Ibler})}{(\text{N.UL.SCH.HalfPiBPSK.TB} + \text{N.UL.SCH.QPSK.TB} + \text{N.UL.SCH.16QAM.TB} + \text{N.UL.SCH.64QAM.TB} + \text{N.UL.SCH.256QAM.TB})} \times 100\%$
Uplink RBLER	$\frac{(\text{N.UL.SCH.HalfPiBPSK.ErrTB.Rbler} + \text{N.UL.SCH.QPSK.ErrTB.Rbler} + \text{N.UL.SCH.16QAM.ErrTB.Rbler} + \text{N.UL.SCH.64QAM.ErrTB.Rbler} + \text{N.UL.SCH.256QAM.ErrTB.Rbler})}{(\text{N.UL.SCH.HalfPiBPSK.TB} + \text{N.UL.SCH.QPSK.TB} + \text{N.UL.SCH.16QAM.TB} + \text{N.UL.SCH.64QAM.TB} + \text{N.UL.SCH.256QAM.TB})} \times 100\%$
Proportion of uplink small packets	$(\text{N.ThpVol.UE.UL.SmallPkt} / \text{N.ThpVol.UL}) \times 100\%$

The uplink service differentiation guarantee function has the following impacts:

- After uplink differentiated service guarantee takes effect, uplink scheduling resources are preempted by guaranteed UEs, which decreases the uplink traffic volumes of common UEs.
- In uplink heavy-load scenarios ($(\text{N.PRB.PUSCH.Used.Avg} / \text{N.PRB.UL.Avail.Avg}) \times 100\% > 60\%$), the average uplink UE throughput in a cell indicated by **User Uplink Average Throughput (DU)** may fluctuate, and the uplink user-perceived rates of guaranteed UEs fluctuate. The specific fluctuation depends on the UE distribution.

The differentiated uplink MU-MIMO scheduling function has the following impacts:

- After differentiated uplink MU-MIMO scheduling takes effect, uplink scheduling resources are preempted by guaranteed UEs, which decreases the uplink traffic volumes of common UEs.
- The average uplink UE throughput in a cell indicated by **User Uplink Average Throughput (DU)** may fluctuate, and the uplink user-perceived rates of guaranteed UEs fluctuate. The specific fluctuation depends on the UE distribution.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-091253	Uplink Service Guarantee	NR0S00ULSG00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

- Uplink live webcast experience guarantee
None
- Uplink differentiated service guarantee

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellCnCovAlgo.UICoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	The uplink differentiated service guarantee function takes effect only when the turbo uplink function is enabled.
Low-frequency TDD	SR-based scheduling optimization	SR_SCH_OPT_SW option of the NRDUCellPusch.PuschPerformanceSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	The uplink differentiated service guarantee function takes effect only when SR-based scheduling optimization is enabled.

- Differentiated uplink MU-MIMO scheduling

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink scheduling enhancement	UL_SCHEDULE_ENH_SW option of the NRDUCellPusch.UIPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	"RB policy for large-packet scheduling in uplink MU-MIMO pairing" takes effect only when uplink scheduling enhancement is enabled.
Low-frequency TDD	Uplink MU grouping and pairing	UL_MU_GRP_PAIR_SW option of the NRDUCellUIMimo.UIMuMimoAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	"RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing" takes effect only when uplink MU grouping and pairing is enabled.

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

- Uplink live webcast experience guarantee

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Stable Video Surveillance Experience	STABLE_UL_SERVICE_EXP_BASIC_SW option of the NRDUCellFeatureSw.StableUIServiceExpSwitch parameter	<i>Stable Video Surveillance Experience</i>	When both uplink live webcast experience guarantee and Stable Video Experience Surveillance are enabled, the number of video surveillance UEs for which stable uplink service experience is guaranteed may be affected.
Low-frequency TDD	VoNR	VONR_SW option of the NRCeIIAlgoSwitch.VonrSwitch parameter	<i>VoNR</i>	Uplink live webcast experience guarantee does not take effect for UEs with VoNR in effect.

- Uplink differentiated service guarantee/Differentiated uplink MU-MIMO scheduling

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

- Uplink live webcast experience guarantee
Table 5-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-3 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	QCI-based Diff Service Experience Switch	NRDUCellFeaturesSw.QciBasedDiffServiceExpSw	To enable the uplink live webcast experience guarantee function, select the UL_LIVE_WEBCAST_EXP_GUAR_SW option.
Low-frequency TDD	Service Type	gNBQciBearer.ServiceType	To enable the uplink live webcast experience guarantee function, set this parameter to UL_LIVE_WEBCAST_SERVICE .
Low-frequency TDD	Stable Uplink Experience Guaranteed Rate	gNBDUSchParamGroup.StableULExpGuarRate	Set this parameter based on the uplink rate requirements and mainstream video resolutions of the live streaming UEs on the network.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Stable Uplink Experience Policy	gNBDUSchParamGroup.StableUExpPol	Set this parameter based on customer policies: - It is recommended that this parameter be set to EXPERIENCE_FIRST if the rates of some uplink live streaming UEs need to be preferentially guaranteed. - It is recommended that this parameter be set to USER_NUM_FIRST when the capacity in terms of the number of uplink live streaming UEs needs to be preferentially guaranteed. - It is recommended that this parameter be set to NEUTRAL when both the rates and the capacity need to be guaranteed.
Low-frequency TDD	Stable UL Service Exp Max UE Num	NRDUCellULSch.StableUExpMaxUeNum	Configure this parameter based on the required number of UEs.
Low-frequency TDD	UL Live Webcast Far UE SINR Threshold	gNBDUSchParamGroup.UILiveWebcastFarUeSinrThld	Use the recommended value.
Low-frequency TDD	UL Live Webcast Measure Traffic Threshold	gNBQciBearer.UILiveWebcastMeasTrfThld	Set this parameter based on the second-level uplink traffic, uplink rate requirements, and mainstream resolutions of live streaming UEs on the network.

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	Priority Level	gNBQciBearer.PriortyLevel	To enable the uplink live webcast experience guarantee function, ensure that the priority of bearers for uplink live streaming services is only lower than that of QCI 1 and higher than that of other bearers.

- Uplink differentiated service guarantee

Table 5-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-4 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	QCI-based Diff Service Experience Switch	NRDUCellFeaturesSw.QciBasedDiffServExpSw	To enable the uplink differentiated service guarantee function, select the UL_SERV_DIFF_EXP_GUAR_SW option.
Low-frequency TDD	Service Differentiation Algorithm Switch	NRDUCellQciBearer.ServiceDiffAlgorithmSwitch	To enable uplink differentiated service guarantee, select the SERVICE_DIFF_GUARANTEE_UE_FLAG option.
Low-frequency TDD	SR Schedule Optimize CCE Fail Rate Threshold	NRDUCellPusch.SrSchOptCceFailRateThld	Use the recommended value.
Low-frequency TDD	SR Schedule Optimize PRB Low Threshold	NRDUCellPusch.SrSchOptPrbLowThld	Use the recommended value. The value of this parameter must be less than or equal to the value of the NRDUCellPusch.SrSchOptPrbHighThld parameter.
Low-frequency TDD	SR Schedule Optimize PRB High Threshold	NRDUCellPusch.SrSchOptPrbHighThld	Use the recommended value.

- Differentiated uplink MU-MIMO scheduling

Table 5-5 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-5 Parameters used for activation

RAT	Parameter Name	Parameter ID	Setting Notes
Low-frequency TDD	QCI-based Diff Service Experience Switch	NRDUCellFeaturesw.QciBasedDiffServExpSw	To enable the differentiated uplink MU-MIMO scheduling function, select the UL_SERV_DIFF_EXP_GUAR_SW option.
Low-frequency TDD	Service Differentiation Algorithm Switch	NRDUCellQciBearer.ServiceDiffAlgorithmSwitch	To enable "RB policy for large-packet scheduling in uplink MU-MIMO pairing", select the UL_MU_BIG_PKT_PAIR_RB_POLI_SW option.
Low-frequency TDD	UL MU-MIMO Algorithm Switch	NRDUCellULMimo.ULMuMimoAlgorithmSwitch	To enable "RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing", select the UL_MU_GRP_PAIR_SR_RB_CTRL_SW option. Before enabling this function, select the UL_MU_GRP_PAIR_RB_TILT_SW option of the NRDUCellQciBearer.ServiceDiffAlgorithmSwitch parameter.
Low-frequency TDD	Group Pair SR RB Control CCE Fail Threshold	NRDUCellULMimo.GroupPairSrRbCtrlCceFailThld	Use the recommended value. A smaller value of this parameter results in a lower probability of "RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing" taking effect.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Enabling uplink live webcast experience guarantee**

```
//Enabling the uplink live webcast experience guarantee function
MOD NRDUCELLFEATURESW: NrDuCellId=1,
QciBasedDiffServExpSw=UL_LIVE_WEBCAST_EXP_GUAR_SW-1;

//Adding a bearer for uplink live streaming services
ADD GNB5QICONFIG: Nr5qi=72, Qci=72;
ADD GNBQCIBEARER: Qci=72;

//Setting the service type corresponding to the QCI to uplink live streaming services for all cells served
by the base station
MOD GNBQCIBEARER: Qci=72, ServiceType=UL_LIVE_WEBCAST_SERVICE;

//Setting the bearer priority of live streaming services lower than that of QCI 1 and higher than those
of other QCIs. For example, if the bearer priorities of QCIs 1 and 5 are 10 and 20, respectively, the
bearer priority of live streaming services needs to be set to a value less than 20 but greater than 10.
MOD GNBQCIBEARER: Qci=72, PriorityLevel=15;

//Configuring a bearer for uplink live streaming services and associating the bearer with a QCI
ADD NRCELLQCIBEARER: NrCellId=1, Qci=72;
ADD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=72;
ADD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=72, ScheduleParamGroupId=1;

//Configuring the maximum number of UEs with stable uplink service experience
MOD NRDUCELLULSCH: NrDuCellId=1, StableUExpMaxUeNum=10;

//For uplink live streaming UEs: Configuring the guaranteed rate, policy, and SINR threshold for
identifying cell-edge UEs running uplink live streaming services
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, StableUExpGuarRate=5000,
StableUExpPol=USER_NUM_FIRST, ULLiveWebcastFarUeSinrThld=50;

//Triggering RRC connection reconfiguration after the parameters of uplink live webcast experience
guarantee are modified for online UEs (that is, after the uplink live streaming service bearer and the
corresponding QCI configuration are modified)
STR NRCELLSERVICEPARAMEFFECT: NrCellId=1, NrNetworkingOption=SA, First5qi=72;
```
- Enabling uplink differentiated service guarantee**

```
//Turning on the UL-Service-Diff Experience Guarantee Sw
MOD NRDUCELLFEATURESW: NrDuCellId=1,
QciBasedDiffServExpSw=UL_SERV_DIFF_EXP_GUAR_SW-1;
//Enabling uplink differentiated service guarantee
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=72,
ServiceDiffAlgoSwitch=SERVICE_DIFF_GUARANTEE_UE_FLAG-1;
//Configuring the CCE failure rate threshold, lower PRB threshold, and upper PRB threshold for SR-
based scheduling optimization
MOD NRDUCELLPUSCH: NrDuCellId=1, SrSchOptCceFailRateThld=30, SrSchOptPrbLowThld=40,
SrSchOptPrbHighThld=90;
```
- Enabling differentiated uplink MU-MIMO scheduling**

```
//Turning on the UL-Service-Diff Experience Guarantee Sw
MOD NRDUCELLFEATURESW: NrDuCellId=1,
QciBasedDiffServExpSw=UL_SERV_DIFF_EXP_GUAR_SW-1;
//Enabling RB policy for large-packet scheduling in uplink MU-MIMO pairing
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=72,
ServiceDiffAlgoSwitch=UL_MU_BIG_PKT_PAIR_RB_POLI_SW-1;
//Enabling RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing
```

```
MOD NRDUCELLULMIMO: NrDuCellId=1,  
ULMuMimoAlgoSwitch=UL_MU_GRP_PAIR_SR_RB_CTRL_SW-1, GrpPairSrRbCtrlCceFailThld=30;  
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=72,  
ServiceDiffAlgoSwitch=UL_MU_GRP_PAIR_RB_TILT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Disabling uplink live webcast experience guarantee
//Disabling the uplink live webcast experience guarantee function
MOD NRDUCELLFEATURESW: NrDuCellId=1,
QciBasedDiffServExpSw=UL_LIVE_WEBCAST_EXP_GUAR_SW-0;
- Disabling uplink differentiated service guarantee
//Disabling uplink differentiated service guarantee
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=72,
ServiceDiffAlgoSwitch=SERVICE_DIFF_GUARANTEE_UE_FLAG-0;
//Turning off the UL-Service-Diff Experience Guarantee Sw
MOD NRDUCELLFEATURESW: NrDuCellId=1,
QciBasedDiffServExpSw=UL_SERV_DIFF_EXP_GUAR_SW-0;
- Disabling differentiated uplink MU-MIMO scheduling
//Disabling RB policy for large-packet scheduling in uplink MU-MIMO pairing
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=72,
ServiceDiffAlgoSwitch=UL_MU_BIG_PKT_PAIR_RB_POLI_SW-0;
//Disabling RB control for SR-based scheduling in uplink MU-MIMO grouping and pairing
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=72,
ServiceDiffAlgoSwitch=UL_MU_GRP_PAIR_RB_TILT_SW-0;
MOD NRDUCELLULMIMO: NrDuCellId=1,
ULMuMimoAlgoSwitch=UL_MU_GRP_PAIR_SR_RB_CTRL_SW-0;
//Turning off the UL-Service-Diff Experience Guarantee Sw
MOD NRDUCELLFEATURESW: NrDuCellId=1,
QciBasedDiffServExpSw=UL_SERV_DIFF_EXP_GUAR_SW-0;

5.4.2 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3 Activation Verification

- Uplink live webcast experience guarantee
After this function is enabled, you can check whether this function has taken effect by observing the value of **N.ThpVol.UL.DuCell5QI** in SA networking or the value of **N.ThpVol.UL.QCI** in NSA networking.
- Uplink differentiated service guarantee
In NSA networking, observe the average number of active UEs (**N.User.RRCCConn.Active.QCI.Avg**) and the maximum number of active UEs (**N.User.RRCCConn.Active.QCI.Max**) of a QCI-specific bearer. If the value of either counter is not 0, this function has taken effect.
In SA networking, flexible performance counter objects need to be configured. In this case, observe the average number of active UEs (**N.User.RRCCConn.Active.Flex.Avg**) and the maximum number of active UEs

(**N.User.RRCCConn.Active.Flex.Max**) of those objects, which indicate the average number of active UEs or the maximum number of active UEs on equivalent QCI. If the value of either counter is not 0, this function has taken effect. For details about how to configure flexible performance counter objects, see "SAI-based Service Experience Improvement" in *SAI-based Service Experience Improvement*.

- Differentiated uplink MU-MIMO scheduling

In NSA networking, observe the average number of active UEs (**N.User.RRCCConn.Active.QCI.Avg**) and the maximum number of active UEs (**N.User.RRCCConn.Active.QCI.Max**) of a QCI-specific bearer. If the value of either counter is not 0 and the QCI-specific average uplink rate, the number of RBs paired for uplink MU-MIMO in a cell, and the average number of layers paired for uplink MU-MIMO increase, this function has taken effect.

In SA networking, observe the average number of active UEs (**N.User.RRCCConn.Active.DuCell5QI.Avg**) and the maximum number of active UEs (**N.User.RRCCConn.Active.DuCell5QI.Max**) of a 5QI-specific bearer. If the value of either counter is not 0 and the 5QI-specific average uplink rate, the number of RBs paired for uplink MU-MIMO in a cell, and the average number of layers paired for uplink MU-MIMO increase, this function has taken effect.

5.4.4 Network Monitoring

The indicators listed in [5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

1. 3GPP TS 38.104: "Base Station (BS) radio transmission and reception (Release 17)"
2. 3GPP TS 38.306: "User Equipment (UE) radio access capabilities (Release 16)"
3. *3900 & 5900 Series Base Station Product Documentation*
4. *UE Power Saving*
5. *Multi-Frequency Smart Selection*
6. *Super Uplink*
7. *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
8. *VoNR*
9. *FWA*
10. *Network Slicing*
11. *License Control Item Lists*
12. *RedCap*
13. *QoS Management*
14. *Service-differentiated Experience Guarantee*
15. *Stable Video Surveillance Experience*
16. *Turbo Uplink (Low-Frequency TDD)*
17. *SAI-based Service Experience Improvement*
18. *Uplink Scheduling*
19. *UE Power Saving*